CAPSTONE PROJECT WRITING GUIDE

A Student Handout for Chapter Content and Writing Conventions

(APA 7th Edition Compliant)

This chapter structure follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework. APA 7th Edition governs writing style and formatting, not project structure.

CHAPTER DESCRIPTIONS

CHAPTER 1: INTRODUCTION

This chapter sets the stage for the entire project. Include the background of the study, problem statement, aim and objectives, research questions (if applicable), significance of the study, scope and limitations, definition of key terms, and organization of the report.

Key Question: "What is the problem, and why does it need to be solved?"

Tense: Use present tense for background facts, present perfect for research trends, and past tense for what was done.

Example: "Machine learning is a rapidly growing field. Researchers have increasingly explored its applications in healthcare. This study aimed to develop a predictive model for..."

CHAPTER 2: REVIEW OF RELATED WORKS

This chapter demonstrates your understanding of existing knowledge. Include the theoretical or conceptual framework, review of previous studies, comparison of related works (a summary table is recommended but not required), the research gap your project addresses, and a summary of the chapter.

Key Question: "What has already been done, and what gap does my project fill?"

Tense: Use past tense for specific author findings, present perfect for summarizing a body of research, and present tense for general facts.

Example: "Ahmed (2021) found that Random Forest outperformed Decision Trees. Several researchers have explored the use of deep learning for sentiment analysis. Machine learning is a subset of artificial intelligence."

CHAPTER 3: METHODOLOGY

This chapter describes the overall approach and framework guiding the project. Include the research design, the framework or process model adopted and justification for choosing it, overview of tools and technologies used, description of algorithms or techniques selected and why, description of data sources, evaluation metrics to be used and why, and any ethical considerations.

Key Question: "What is my game plan and why?"

Tense: Use past tense for what was done and present perfect for established practices.

Example: "The CRISP-DM framework was adopted for this study. CRISP-DM has been widely used in data mining projects (Wirth and Hipp, 2000). Python was selected as the primary programming language because it provides extensive libraries for machine learning."

CHAPTER 4: DATA UNDERSTANDING

This chapter explores and describes the raw data before any changes are made. Include a description of the dataset (number of records, features, data types), summary statistics (mean, median, mode, standard deviation, min, max), exploratory data analysis with visualizations (histograms, box plots, scatter plots, heatmaps, correlation matrices), data quality assessment (missing values, outliers, duplicates, inconsistencies), and initial observations and insights.

Key Question: "What does my raw data look like, and what did I discover about it?"

Tense: Use past tense for findings and present tense when referring to tables and figures.

Example: "The dataset contained 5,000 records and 15 features. As shown in Table 4.1, the target variable was highly imbalanced. Figure 4.2 illustrates the distribution of the age feature."

Note: APA 7th edition requires tables and figures to be numbered consecutively throughout the entire manuscript (Table 1, Table 2, Figure 1, Figure 2, etc.), regardless of chapter. Some institutions adapt APA by numbering per chapter (e.g., Table 4.1 for the first table in Chapter 4). For our program, follow APA's continuous numbering for consistency.

CHAPTER 5: DATA PREPARATION

This chapter documents all preprocessing and transformation steps. Include how missing values were handled (deletion, imputation) with justification, how outliers were handled with justification, encoding of categorical variables, feature scaling or normalization, feature engineering, feature selection, handling of class imbalance (if applicable), data splitting into training and test sets, and a description of the final cleaned dataset.

Key Question: "What did I do to clean and transform the data, and why?"

Tense: Use past tense throughout.

Example: "Missing values in the income feature were imputed using the median. Categorical variables were encoded using one-hot encoding. The dataset was split into 80% training and 20% testing sets."

CHAPTER 6: MODELING

This chapter focuses on the construction and training of the models. Include the models implemented with detailed descriptions, model architecture and configuration (hyperparameters, settings, network architecture for deep learning), hyperparameter tuning methods and results, the training process and any challenges encountered, tools and libraries used, and code snippets or screenshots if required.

Key Question: "How exactly did I build and train my model(s)?"

Tense: Use past tense throughout.

Example: "A Random Forest classifier was implemented using the scikit-learn library. The number of estimators was set to 100. Hyperparameter tuning was performed using Grid Search with 5-fold cross-validation."

CHAPTER 7: EVALUATION

This chapter assesses how well the models performed. Include evaluation metric results for each model (accuracy, precision, recall, F1-score, AUC-ROC, confusion matrix for classification; MAE, MSE, RMSE, R-squared for regression), comparison of models using tables and charts, confusion matrix analysis, cross-validation results, overfitting and underfitting analysis, best model selection with justification, and assessment against the objectives from Chapter 1.

Key Question: "How good is my model, and how do I know?"

Tense: Use past tense for results and present tense for tables and figures.

Example: "The Random Forest classifier achieved an F1-score of 0.92. As shown in Table 7.1, this model outperformed all other classifiers. Figure 7.3 presents the ROC curves for all models."

CHAPTER 8: RESULTS AND DISCUSSION

This chapter ties findings back to the bigger picture. Include a summary of key results, interpretation of results in context, comparison with related works from Chapter 2, implications of the findings, unexpected findings and possible explanations, and limitations encountered.

Key Question: "What do my results mean, and how do they compare to existing work?"

Tense: Use past tense for findings and present tense for interpretations and implications.

Example: "The Random Forest model outperformed the other classifiers. This result is consistent with the findings of Ahmed (2021). The results suggest that ensemble methods are more suitable for imbalanced datasets."

CHAPTER 9: CONCLUSIONS

This chapter provides a concise wrap-up. Include a summary of the entire project, whether each objective was met, key contributions, and final remarks.

Key Question: "Did I achieve what I set out to do?"

Tense: Use past tense for summarizing and present tense for contributions and implications.

Example: "This study aimed to develop a predictive model for customer churn. All three objectives were achieved. The findings contribute to the growing body of knowledge on churn prediction."

CHAPTER 10: RECOMMENDATIONS

This chapter looks forward. Include recommendations for future work, suggestions for practitioners, areas for further research, and technical improvements.

Key Question: "What should be done next?"

Tense: Use present tense and modal verbs (could, should, may).

Example: "Future researchers could explore the use of deep learning models. It is recommended that the model be deployed as a web-based application."

APA 7TH EDITION VERB TENSE GUIDELINES (Section 4.12)

PAST TENSE — Use when describing specific actions, procedures, and results.
Examples: "Smith (2020) found that..." / "The data was collected from..." / "The model achieved 95% accuracy."

PRESENT PERFECT TENSE — Use when summarizing a body of research or indicating something that began in the past and continues.
Examples: "Researchers have shown that..." / "Several studies have demonstrated that..." / "Deep learning has become increasingly popular."

PRESENT TENSE — Use for general truths, established knowledge, referring to tables and figures, and discussing implications.
Examples: "The results suggest that..." / "Decision Trees are supervised algorithms." / "Table 4.1 shows the distribution."

COMMON MISTAKE: PROPOSAL VS. FINAL PAPER

Proposal language (INCORRECT for final paper): "The data will be collected from..."
Final paper language (CORRECT): "The data was collected from..."

Proposal language (INCORRECT): "SMOTE will be applied to..."
Final paper language (CORRECT): "SMOTE was applied to..."

If your final paper still reads like a proposal, you need to revise it.

TENSE SUMMARY BY CHAPTER

Chapter 1 (Introduction): Present, Present Perfect, and Past
Chapter 2 (Related Works): Past, Present Perfect, and Present
Chapter 3 (Methodology): Past and Present Perfect
Chapter 4 (Data Understanding): Past and Present (for figures and tables)
Chapter 5 (Data Preparation): Past
Chapter 6 (Modeling): Past
Chapter 7 (Evaluation): Past and Present (for figures and tables)
Chapter 8 (Results and Discussion): Past and Present
Chapter 9 (Conclusions): Past and Present
Chapter 10 (Recommendations): Present and Modal Verbs

KEY APA 7TH FORMATTING REMINDERS

Font: 12-pt Times New Roman, 11-pt Calibri, or 11-pt Arial
Margins: 1.5 inch on left side, 1 inch (2.54 cm) on other sides
Spacing: Double-spaced throughout
Alignment: Left-aligned
Paragraph Indent: 0.5 inch first-line indent
Page Numbers: Top right corner
In-Text Citations: (Author, Year) or Author (Year)
Three or More Authors: (Smith et al., Year)
Numbers: Spell out below 10, use numerals for 10 and above
References: Hanging indent, double-spaced, alphabetical order